

Kelvin Xu

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Science, GPA: 3.86/4.00

September 2021 – May 2025

EXPERIENCE

Software Engineer Co-op

January 2024 – June 2024

Transperfect Legal Solutions

Maynard, MA

- Collaborated within a small development team to build and enhance an industry-leading e-discovery and early case assessment management platform, leveraging Java, PostgreSQL, and Hibernate for backend development.
- Redesigned the import/export process for stored client project data in the application, improving compatibility across devices and enabling seamless migration to newer versions while increasing efficiency and speed.
- Initiated and prototyped integration of the DocuSign API, enabling clients to manage and sign documents directly within the application without relying on external software.

Quality Assurance Engineer Co-op

January 2023 – August 2023

Transperfect Legal Solutions

Maynard, MA

- Developed automated front-end test suites in Java using Selenium and JUnit to detect bugs and validate performance across multiple interfaces, applying object-oriented design principles to ensure modular code.
- Resolved 15+ development tickets, collaborating closely with team members through peer reviews and clear communication, and producing thorough documentation to support implementation and future maintenance.

IT Assistant

September 2021– December 2022

Northeastern University

Boston, MA

- Provided technical support by diagnosing and resolving hardware, software, and connectivity issues, while facilitating communication between manager, faculty, and students to ensure smooth resolution.
- Installed, configured, and maintained computers, printers, and peripheral devices, ensuring smooth operation and providing clear setup guidance.
- Managed and organized a comprehensive inventory of computer components, streamlining hardware replenishment and contributing to improved workplace productivity.

PROJECTS

OpenGL Rendering Engine | C++, OpenGL, CMake, Git

March 2025 – April 2025

- Created a real-time graphics engine supporting both rasterization and ray tracing using OpenGL and C++.
- Implemented a modular scenegraph architecture to manage hierarchical transformations, define light sources, and enable keyframe-based animations with interactive scene rotation for enhanced visualization and control.
- Added support for texture mapping and bump mapping for enhanced surface detail and graphical fidelity.

Spreadsheet Application | TypeScript, React, Express.js, Jest, Git

November 2024 – December 2024

- Developed a TypeScript-based spreadsheet application with a React front-end, enabling core functionalities like cell editing, complex formula calculations, cell referencing, row and column resizing, and error handling.
- Implemented advanced features, including customizable graph creation for various graph types, a custom formula builder, cell history, and version history, enhancing user flexibility and workflow.
- Followed Test-Driven Development and Object-Oriented Design principles, utilizing design patterns such as the visitor pattern, singletons, and generics for maintainability and scalability.

Food Delivery Platform | Python, Flask, MySQL, Ngrok, AppSmith

October 2024 – November 2024

- Designed and implemented an online food delivery service in Python, modeling system interactions with UML to support order management and driver dispatch.
- Extended the system with a front-end AppSmith application, deployed using Docker containers with a Flask REST API and MySQL database, with Ngrok enabling seamless client-to-API communication.

TECHNICAL SKILLS

Languages: Java, Python, C#, C/C++, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks: React, Node.js, Express.js, Flask, Spring Boot, Hibernate

Developer Tools: Git, Docker, IntelliJ, Unity, VS Code, CMake, BitBucket, JIRA, Confluence

Libraries: OpenGL, JUnit, Mockito, Jest, NumPy, Pandas, Newtonsoft.Json

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10/23/25

Michigan Engineering Robotics Department

University of Michigan

Dear Hiring Team,

I am excited to apply for the Software Developer position in the Robotics Department at Michigan Engineering. As a graduate from Northeastern University with a Bachelor of Science in Computer Science degree, I bring hands-on experience in software development, IT support, and testing that aligns closely with your department's mission to build a people-first technological future.

At TransPerfect Legal Solutions, I contributed to developing an e-discovery platform using Java, PostgreSQL, and Hibernate. I improved data compatibility and efficiency by redesigning the import/export process and prototyped a DocuSign API integration to streamline client workflows. My previous QA Co-op experience involved automating test suites in Java with Selenium, strengthening my debugging and documentation skills. At the university, as an IT Assistant, I provided direct technical support to faculty and students. I resolved hardware, software, and connectivity issues with clear, user-focused communication.

Beyond work, I have built several full-stack applications, including a React-based spreadsheet tool and a Flask/MySQL delivery platform, applying design patterns, database scripting, and test-driven development to ensure reliability and scalability.

I am drawn to Michigan Engineering's focus on innovation, inclusion, and community service. I would be honored to contribute my technical and collaborative skills toward developing solutions that support the Robotics Department's mission.

Thank you for your consideration.

Sincerely,

Kelvin Xu